

TAMIL NADU STATE BOARD - STANDARD 11 PHYSICS VOLUME 1

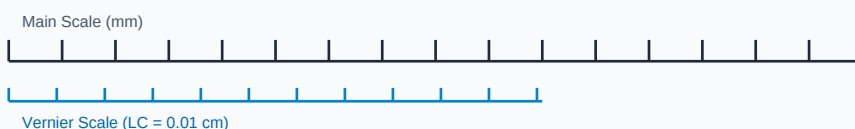
CHAPTER 4: WORK, ENERGY AND POWER

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Physics Volume 1
Chapter Name:	Work, Energy and Power

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

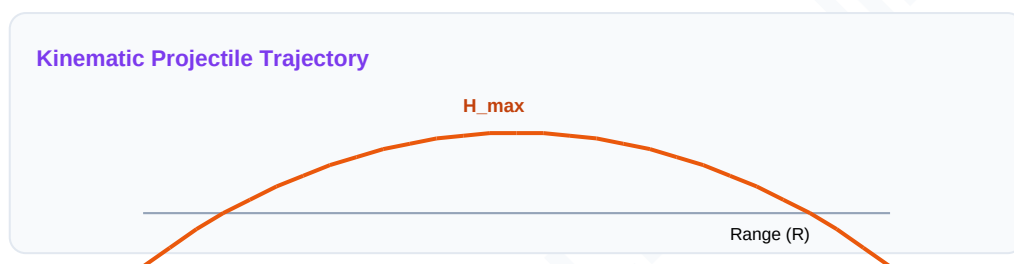
Measurement Linear Scale Division

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Work Done	The product of the component of force along the direction of displacement and the magnitude of displacement.
Conservative Force	A force where the work done in moving a particle between two points is independent of the path taken.
Coefficient of Restitution	The ratio of relative velocity of separation to relative velocity of approach after a collision.

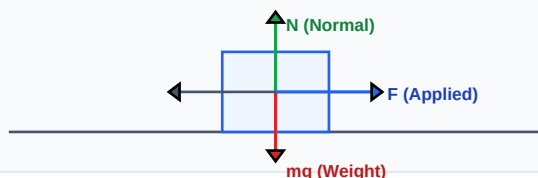
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Work: $W = F \cdot d = F \cdot d \cdot \cos(\theta)$
- Kinetic Energy: $KE = 0.5 \cdot m \cdot v^2$ | Potential: $PE = m \cdot g \cdot h$
- Power: $P = dW/dt = F \cdot v$
- Coefficient of Restitution: $e = (v_2 - v_1) / (u_1 - u_2)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

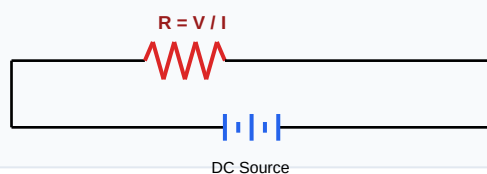
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Calculate the work done in lifting a 10kg block to a height of 5m. ($g = 9.8 \text{ m/s}^2$)

Step-by-step Solution:

$$W = F \cdot d = m \cdot g \cdot h = 10 \cdot 9.8 \cdot 5 = 490 \text{ Joules.}$$

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Show that for a perfectly elastic 1D collision, the relative velocity of approach equals relative velocity of separation.

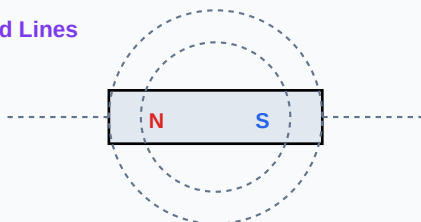
Analytical Solution Strategy:

Conservation of momentum: $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$.

Conservation of KE: $0.5 m_1 u_1^2 + 0.5 m_2 u_2^2 = 0.5 m_1 v_1^2 + 0.5 m_2 v_2^2$.

Dividing the KE expression by momentum expression yields: $u_1 - u_2 = v_2 - v_1$.

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State the work-energy theorem.

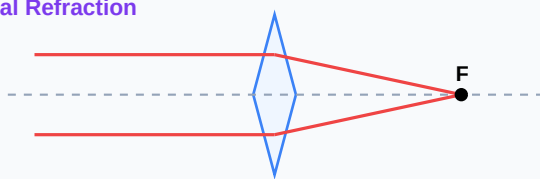
Q2 (2-Mark): A 20W bulb is turned on for 10 seconds. Find the energy consumed.

Q3 (2-Mark): Differentiate between elastic and inelastic collisions.

Q4 (5-Mark): What is potential energy of a stretched spring? Write its expression.

Q5 (5-Mark): Show that mechanical energy is conserved during free fall.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Hydropower Plants

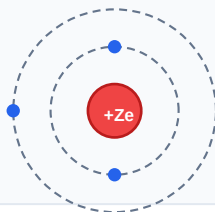
Gravitational potential energy of dammed water is converted to kinetic energy to rotate power turbines.

Application Case: Sports Technology

Golf balls are engineered with high coefficients of restitution to travel further upon impact.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Sinusoidal AC Voltage Waveform

